

Machine Intelligence

3. Reasoning under Uncertainty, Bayesian Networks

(Our Machinery for) Thinking About What is Likely to be True

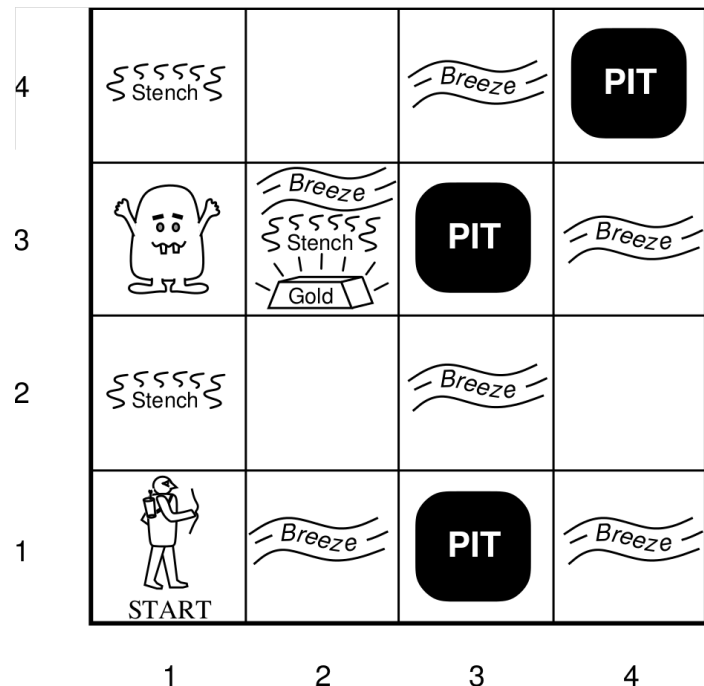
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Fall 2024

The Wumpus World



- **Actions:** *GoForward*, *TurnRight* (by 90°), *TurnLeft* (by 90°), *Grab* object in current cell, *Shoot* arrow in direction you're facing (you got exactly one arrow), *Leave cave* if you're in cell [1,1].

→ **Fall down Pit, meet live Wumpus: Game Over.**

- **Initial knowledge:** You're in cell [1,1] facing east. There's a Wumpus, and there's gold.
- **Goal:** Have the gold and be outside the cave.

Percepts: [*Stench*, *Breeze*, *Glitter*, *Bump*, *Scream*]

- Cell adjacent (i.e. north, south, west, east) to Wumpus: *Stench* (else: *None*).
- Cell adjacent to Pit: *Breeze* (else: *None*).
- Cell that contains gold: *Glitter* (else: *None*).
- You walk into a wall: *Bump* (else: *None*).
- Wumpus shot by arrow: *Scream* (else: *None*).

Reasoning in the Wumpus World

A: Agent, **V:** Visited, **OK:** Safe, **P:** Pit, **W:** Wumpus, **B:** Breeze, **S:** Stench, **G:** Gold

1,4	2,4	3,4	4,4
1,3	2,3	3,3	4,3
1,2 OK	2,2	3,2	4,2
1,1 A OK	2,1 OK	3,1	4,1

(1) Initial state

1,4	2,4	3,4	4,4
1,3	2,3	3,3	4,3
1,2 OK	2,2 P?	3,2	4,2
1,1 V OK	2,1 A B OK	3,1 P?	4,1

(2) One step to right

1,4	2,4	3,4	4,4
1,3 W!	2,3	3,3	4,3
1,2 A S OK	2,2 OK	3,2	4,2
1,1 V OK	2,1 B V OK	3,1 P!	4,1

(3) Back, and up to [1,2]

→ **The Wumpus is in [1,3]! How do we know?** Because in [2,1] we perceived no Stench, the Stench in [1,2] can only come from [1,3].

→ **There's a Pit in [3,1]! How do we know?** Because in [1,2] we perceived no Breeze, the Breeze in [2,1] can only come from [3,1].

Uncertainty and Logic

Diagnosis: We want to build an expert dental diagnosis system, that deduces the cause (the disease) from the symptoms.

→ **Can we base this on logic?**

Attempt 1: Say we have a toothache. How's about: $toothache \rightarrow cavity$

→ **Is this rule correct?** No, toothaches may have different causes.

Attempt 2: So what about this: $toothache \rightarrow cavity \vee gum_disease \vee \dots$

→ We don't know all possible causes. And we'd like to be able to deduce which causes are more plausible!

Attempt 3: Perhaps a *causal* rule is better? $cavity \rightarrow toothache$

- **Is this rule correct?** No, not all cavities cause toothaches.
- **Does this rule allow to deduce a cause from a symptom?** No, setting *toothache* to true here has no consequence on the truth of *cavity*. [Note: If *toothache* is *false*, we would conclude $\neg cavity$... which would be incorrect, cf. previous question.]
- Anyway, this still doesn't allow to compare the plausibility of different causes.

→ Logic does not allow to weigh different alternatives, and it does not allow to express incomplete knowledge ("cavity does not always come with a toothache, nor vice versa").

Beliefs and Probabilities

What do we model with probabilities? Incomplete knowledge! We are not 100% sure, but we *believe to a certain degree* that something is true.

→ **Probability $\approx$ Our degree of belief, given our current knowledge.**

Example (Diagnosis)

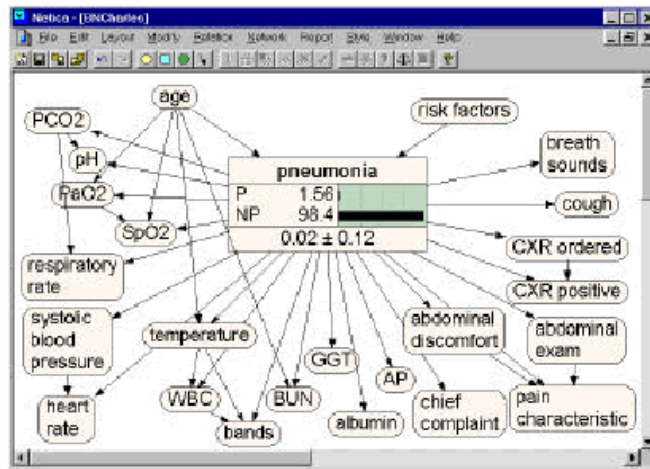
- *toothache* → *cavity* with 80% probability.
- But, for any given p , in reality we do, or do not, have cavity: 1 or 0!
→ The “probability” depends on our knowledge! The “80%” refers to **the fraction of cavity, within the set of all persons that are indistinguishable from the one being evaluated based on our knowledge.**
- If we receive new knowledge (e.g., if we know *gum_disease*) is true, the probability of *cavity* changes!

→ Probabilities represent and measure the uncertainty that stems from lack of knowledge.

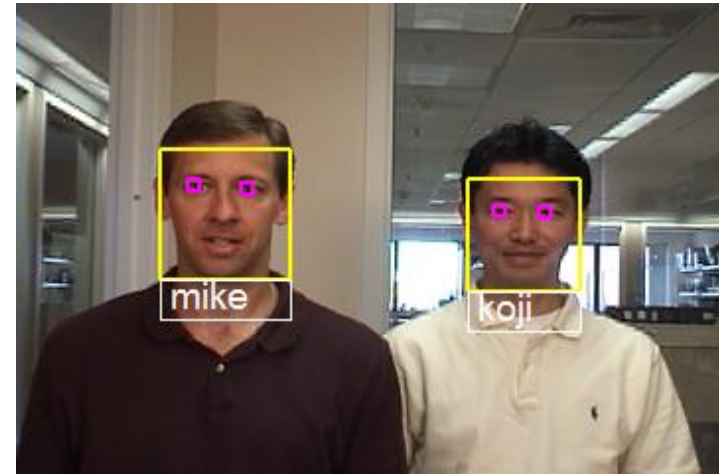
Some Applications

→ A ubiquitous problem: Observe “symptoms”, need to infer “causes”.

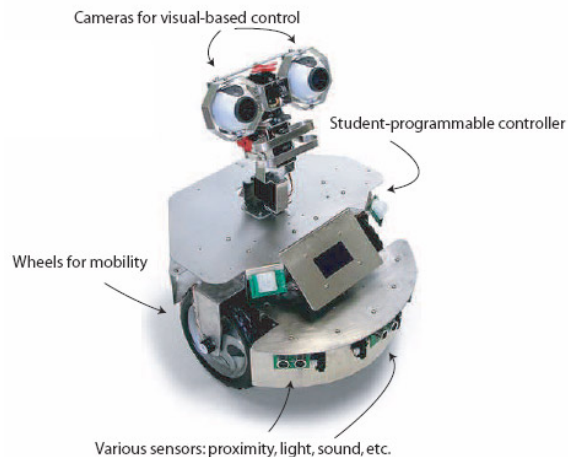
Medical Diagnosis



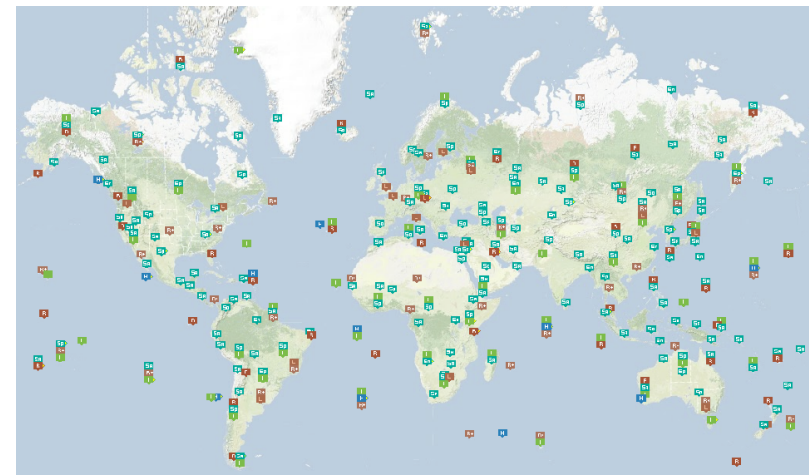
Face Recognition



Self-Localization



Nuclear Test Ban



Our Agenda for This Chapter

- **Quantifying Uncertainty:** What to do when all we know is we don't know.
→ **A bit of motivation for what comes next.**
- **Bayes' Rule:** What's that "Bayes"? How is it used and why is it important?
→ **The basic insight about how to invert the "direction" of conditional probabilities.**
- **Independence:** How to capture and exploit complex relations between random variables?
- **What is a Bayesian Network?**
→ **Tells you what Bayesian networks are.**
- **Computing Probabilities of Atomic Events**
→ **Tells you how to recover the full joint probability distribution out of the BN.**
- **Constructing Bayesian Networks:** How do we design these networks? What effect do our choices have on their size?
→ **Before you can start doing inference, you need to model your domain.**
- **Inference in Bayesian Networks:** **Next Chapter**

Random Variables and Distributions

Definition (Random Variables). Variables defining possible worlds on which probabilities are defined are called **random variables**.

Distributions

For a random variable A , and $a \in D_A$ we have the probability

$$P(A = a) = P(\{\omega \in \Omega \mid A = a \text{ in } \omega\})$$

The **probability distribution of** A (written $P(A)$) is the function on D_A that maps each a to its probability $P(A = a)$.

Example:

$$\mathbf{P}(\textit{Headache}) = \langle F \mapsto 0.1, T \mapsto 0.9 \rangle$$

$$\mathbf{P}(\textit{Weather}) = \langle \textit{sunny} \mapsto 0.7, \textit{rain} \mapsto 0.2, \textit{cloudy} \mapsto 0.08, \textit{snow} \mapsto 0.02 \rangle$$

Joint Probability Distributions

Joint Distributions

Extension to several random variables: $P(A_1, \dots, A_k)$ is the **joint distribution of** $A_1, \dots, A_k$. The joint distribution maps tuples $(a_1, \dots, a_k)$ with $a_i \in D_{A_i}$ to the probability $P(A_1 = a_1, \dots, A_k = a_k)$

Example: $P(\text{Headache}, \text{Weather}) =$

	<i>Headache = true</i>	<i>Headache = false</i>
<i>Weather = sunny</i>	$P(W = \text{sunny} \wedge \text{headache})$	$P(W = \text{sunny} \wedge \neg \text{headache})$
<i>Weather = rain</i>		
<i>Weather = cloudy</i>		
<i>Weather = snow</i>		

Terminology:

- Given random variables $\{X_1, \dots, X_n\}$, an **atomic event (world)** is an assignment of values to all variables.
- Given random variables $\{X_1, \dots, X_n\}$, the **full joint probability distribution**, denoted $P(X_1, \dots, X_n)$, lists the probabilities of all atomic events.

→ The sum of all fields is 1

Probabilities of propositions

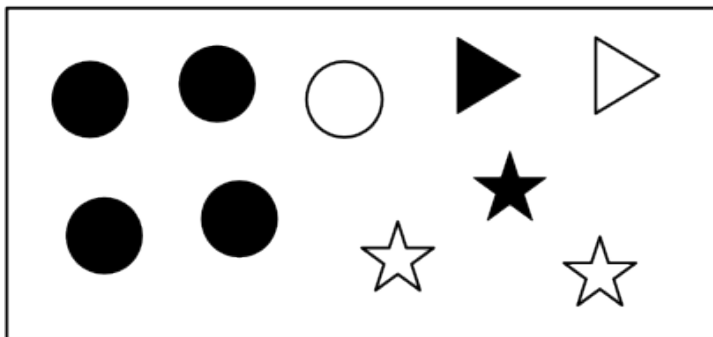
A probability distribution over possible worlds defines probabilities for formulas φ :

$$P(\alpha) = \sum_{\omega \in \Omega: \omega \in M(\varphi)} P(\omega)$$

→ Propositions represent sets of atomic events: the interpretations satisfying the formula.

Notation: Instead of $P(a \wedge b)$, we often write $P(a, b)$.

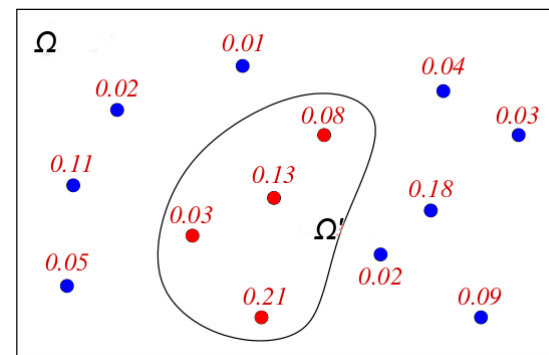
Example



Assume probability for each world is 0.1:

- $P(\text{Shape} = \text{circle}) = 0.5$
- $P(\text{Filled} = \text{false}) = 0.4$
- $P(\text{Shape} = c \wedge \text{Filled} = f) = 0.1$

Another example



$$P(\text{Color} = \text{red}) = 0.08 + 0.13 + 0.03 + 0.21 = 0.45$$

Updating Your Beliefs

→ Do probabilities change as we gather new knowledge?

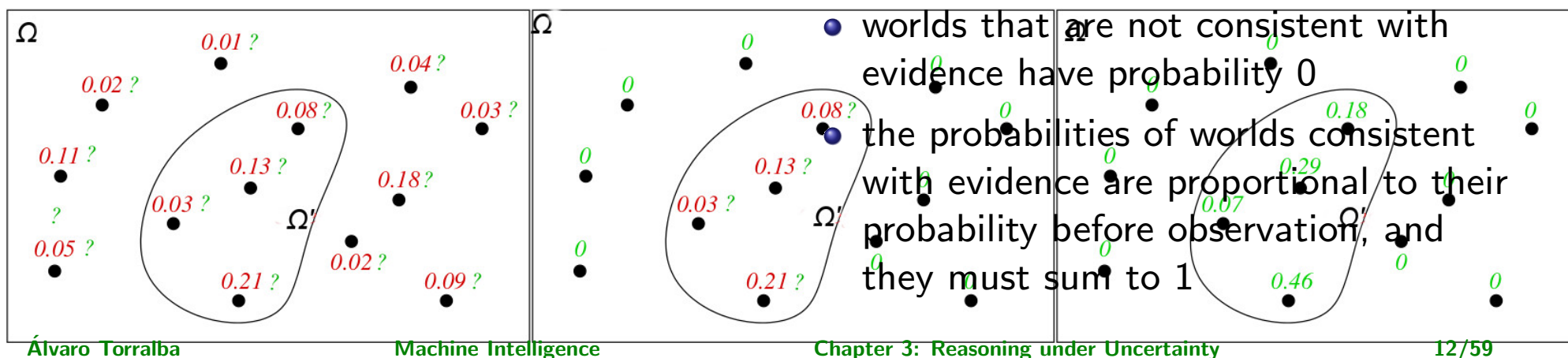
Yes! Probabilities model our *belief*, thus they depend on our knowledge. Given new information (**evidence**), degrees of belief change.

Example: Your “probability of missing the connection train” increases when you are informed that your current train has 30 minutes delay.

Evidence can consist of:

- learning that a certain proposition p is true (“*switch_up*”)
- obtaining partial information on the value of some variable (“ $\text{room_ai} \neq 0.2.90$ ”)
- ...

→ In all cases: **evidence can be represented as the set of possible world Ω' not ruled out by the observation.** How should the probabilities be updated, when I observe Ω' ?



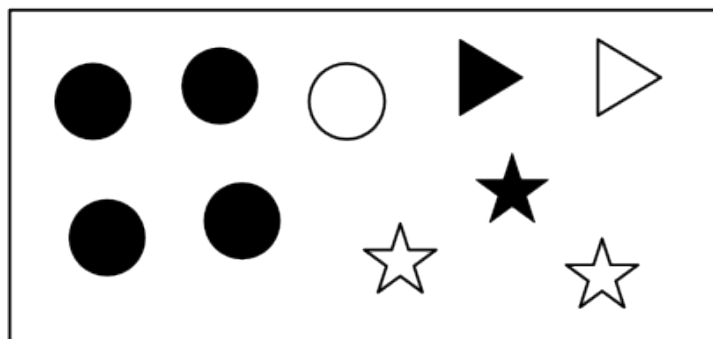
Conditional Probabilities

Definition. Given propositions p and e where $P(e) \neq 0$, the **conditional probability**, or **posterior probability**, of p given e , written $P(p | e)$, is defined as:

$$P(p | e) = \frac{P(p \wedge e)}{P(e)}$$

→ The likelihood of having p **and** e , **within** the set of outcomes where we have e .

Example

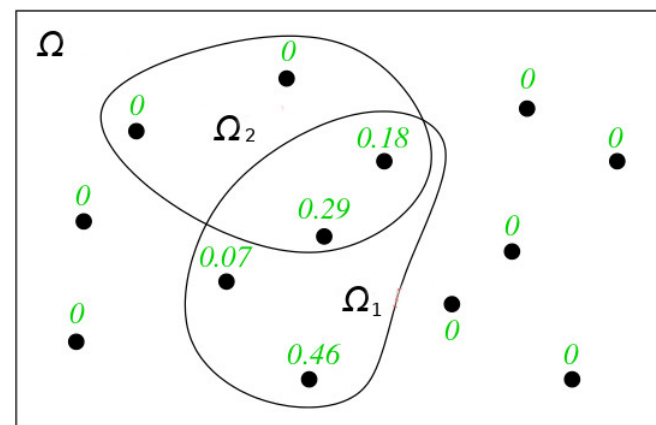


(probability for each world is 0.1)

$$\begin{aligned} P(S=circ. | Fill = f) &= \frac{P(S=circ. \wedge Fill = f)}{P(Fill = f)} \\ &= \frac{0.1}{0.4} = 0.25 \end{aligned}$$

What is the probability of $P(S=star | Fill = f)$?

Another example



- e and p are represented by possible worlds Ω_1 and Ω_2
- division by $P(\Omega_1)$ already in green numbers

$$P(\Omega_2 | \Omega_1) = 0.18 + 0.29$$

Conditional Probability Distributions

Definition. Given random variables X and Y , the **conditional probability distribution** of X given Y , written $\mathbf{P}(X \mid Y)$, is the table of all conditional probabilities of values of X given values of Y .

→ For sets of variables: $\mathbf{P}(X_1, \dots, X_n \mid Y_1, \dots, Y_m)$.

Example: $\mathbf{P}(\text{Weather} \mid \text{Headache}) =$

	<i>Headache = true</i>	<i>Headache = false</i>
<i>Weather = sunny</i>	$P(W = \text{sunny} \mid \text{headache})$	$P(W = \text{sunny} \mid \neg \text{headache})$
<i>Weather = rain</i>		
<i>Weather = cloudy</i>		
<i>Weather = snow</i>		

→ **"The probability of sunshine given that I have a headache?"** If you're susceptible to headaches depending on weather conditions, this makes sense. Otherwise, the two variables are **independent** (we'll get to this later).

Working with the Full Joint Probability Distribution

Example:

	<i>toothache</i>	$\neg$ <i>toothache</i>
<i>cavity</i>	0.12	0.08
$\neg$ <i>cavity</i>	0.08	0.72

→ **How to compute $P(\text{cavity})$?** Sum across the row:

$$P(\text{cavity} \wedge \text{toothache}) + P(\text{cavity} \wedge \neg \text{toothache})$$

→ **How to compute $P(\text{cavity} \vee \text{toothache})$?** Sum across atomic events:

$$P(\text{cavity} \wedge \text{toothache}) + P(\neg \text{cavity} \wedge \text{toothache}) + P(\text{cavity} \wedge \neg \text{toothache})$$

→ **How to compute $P(\text{cavity} \mid \text{toothache})$?** $\frac{P(\text{cavity} \wedge \text{toothache})}{P(\text{toothache})}$

→ All relevant probabilities can be computed using the full joint probability distribution, by expressing propositions as disjunctions of atomic events.

Working with the Full Joint Probability Distribution??

→ **So, is it a good idea to use the full joint probability distribution?** No:

- Given n random variables with k values each, the joint probability distribution contains k^n probabilities.
- Computational cost of dealing with this size.
- *Practically impossible to assess all these probabilities.*

→ **So, is there a compact way to represent the full joint probability distribution?**
Is there an efficient method to work with that representation?

→ Not in general, but it works in many cases. We can work directly with conditional probabilities, and exploit **(conditional) independence**.

→ **Bayesian networks**. (First, we do the simple case.)

Bayes' Rule

Proposition (Bayes' Rule). *Given propositions A and B where $P(a) \neq 0$ and $P(b) \neq 0$, we have:*

$$\mathbf{P(a \mid b) = \frac{P(b \mid a)P(a)}{P(b)}}$$

Proof. By definition, $P(a \mid b) = \frac{P(a \wedge b)}{P(b)}$ which by product rule $P(a \wedge b) = P(b \mid a)P(a)$ is equal to the claim.

Notation: (System of equations)

$$\mathbf{P(X \mid Y) = \frac{P(Y \mid X)P(X)}{P(Y)}}$$

Applying Bayes' Rule

Example: Say we know that $P(\text{toothache} \mid \text{cavity}) = 0.6$, $P(\text{cavity}) = 0.2$, and $P(\text{toothache}) = 0.2$.

→ **We can compute $P(\text{cavity} \mid \text{toothache})$:** By Bayes' rule,
$$P(\text{cavity} \mid \text{toothache}) = \frac{P(\text{toothache} \mid \text{cavity})P(\text{cavity})}{P(\text{toothache})} = \frac{0.6 \cdot 0.2}{0.2} = 0.6.$$

Ok, but: Why don't we simply assess $P(\text{cavity} \mid \text{toothache})$ directly?

- $P(\text{toothache} \mid \text{cavity})$ is **causal**, $P(\text{cavity} \mid \text{toothache})$ is **diagnostic**.
- **Causal dependencies are robust over frequency of the causes.**
→ Example: If there is a cavity epidemic then $P(\text{cavity} \mid \text{toothache})$ increases, but $P(\text{toothache} \mid \text{cavity})$ remains the same.
- Also, causal dependencies are often easier to assess.

→ Bayes' rule allows to perform diagnosis (observing a symptom, what is the cause?) based on prior probabilities and causal dependencies.

Independence

Definition. Events a and b are **independent** if $P(a \wedge b) = P(a)P(b)$.

Proposition. Given independent events a and b where $P(b) \neq 0$, we have $P(a \mid b) = P(a)$.

Proof. By definition, $P(a \mid b) = \frac{P(a \wedge b)}{P(b)}$, which by independence is equal to $\frac{P(a)P(b)}{P(b)} = P(a)$.
($\rightarrow$ Similarly, if $P(a) \neq 0$, we have $P(b \mid a) = P(b)$.)

Examples:

- $P(\text{Dice1} = 6 \wedge \text{Dice2} = 6) = 1/36$.
- $P(W = \text{sunny} \mid \text{headache}) = P(W = \text{sunny})$ unless you're weather-sensitive (cf. slide 26).
- **But toothache and cavity are NOT independent.** The fraction of “cavity” is higher within “toothache” than within “ $\neg$ toothache”. $P(\text{toothache}) = 0.2$ and $P(\text{cavity}) = 0.2$, but $P(\text{toothache} \wedge \text{cavity}) = 0.12 > 0.04$.

Definition. Random variables X and Y are independent if $\mathbf{P}(X, Y) = \mathbf{P}(X)\mathbf{P}(Y)$. (System of equations!)

Example: Football statistics

Results for Bayern Munich and SC Freiburg in seasons 2001/02 and 2003/04. (Not counting the matches Munich vs. Freiburg):

$$D_{Munich) = D_{Freiburg} = \{Win, Draw, Loss\}$$

2001/02

Munich: LWDWWWWWWLDDLWLWDDWWWW

Freiburg: WLLDDWLDWDLWLLDDLWDDLWLLLLLWLV

2003/04

Munich: WDWLDWDLWDDWDLWWDDWWLWLL

Freiburg: LDDWDLWLLLWLVWLLDWLDDWDLWLL

Summary:

Munich	Freiburg			
	W	D	L	
W	12	9	15	36
D	3	4	9	16
L	6	4	2	12
	21	17	26	

Independence of Outcomes

The joint distribution of *Munich* and *Freiburg*:

$P(\text{Munich}, \text{Freiburg})$:

<i>Munich</i>	<i>Freiburg</i>			$P(\text{Munich})$
	W	D	L	
W	.1875 .571	.1406 .529	.2344 .577	.5625
D	.0468 .143	.0625 .235	.1406 .346	
L	.0937 .285	.0625 .235	.0312 .077	.1875
$P(\text{Freiburg})$	.3281	.2656	.4062	

Conditional distribution: $P(\text{Munich} \mid \text{Freiburg})$

We have (almost):

$$P(\text{Munich} \mid \text{Freiburg}) = P(\text{Munich})$$

The variables *Munich* and *Freiburg* are **independent**.

Independent Variables

Definition of Independence

The variables $A_1, \dots, A_k$ and $B_1, \dots, B_m$ are **independent** if

$$P(A_1, \dots, A_k \mid B_1, \dots, B_m) = P(A_1, \dots, A_k)$$

This is equivalent to:

$$P(B_1, \dots, B_m \mid A_1, \dots, A_k) = P(B_1, \dots, B_m)$$

and also to:

$$P(A_1, \dots, A_k, B_1, \dots, B_m) = P(A_1, \dots, A_k) \cdot P(B_1, \dots, B_m)$$

Compact Specifications by Independence

The variables $A_1, \dots, A_k$ and $B_1, \dots, B_m$ are **independent** if

$$P(A_1, \dots, A_k \mid B_1, \dots, B_m) = P(A_1, \dots, A_k)$$

This is equivalent to:

$$P(A_1, \dots, A_k, B_1, \dots, B_m) = P(A_1, \dots, A_k) \cdot P(B_1, \dots, B_m)$$

→ Independence can be exploited to represent the full joint probability distribution more compactly.

$M =$	$F =$			$P(M)$
	W	D	L	
W	M and F are independent			.5625
D				.25
L				.1875
$P(F)$	.3281	.2656	.4062	

The probability for each possible world then is defined, e.g.

$$P(M = D, F = L) = 0.25 \cdot 0.4062 = 0.10155$$

→ Usually, random variables are independent only under particular conditions:

conditional independence, see next section.

Questionnaire

Hair length : $D_{\text{Hair length}} = \{\text{long}, \text{short}\}$

Height : $D_{\text{Height}} = \{\text{tall}, \text{medium}\}$

Sex : $D_{\text{Sex}} = \{\text{male}, \text{female}\}$

Question!

Are Hair length and Height independent?

(A): Yes

(B): No

Joint Distribution:

	Sex			
	male		female	
	<i>Hair length</i>		<i>Hair length</i>	
<i>Height</i>	<i>long</i>	<i>short</i>	<i>long</i>	<i>short</i>
<i>tall</i>	0.06	0.24	0.07	0.03
<i>medium</i>	0.04	0.16	0.28	0.12

$P(\text{Hair length}, \text{Height})$ $P(\text{Height})$,
 $P(\text{Height} \mid \text{Hair length})$:

<i>Height</i>	<i>Hair length</i>		
	<i>long</i>	<i>short</i>	
<i>tall</i>	0.13	0.27	0.4
	0.289	0.49	
<i>medium</i>	0.32	0.28	0.6
	0.711	0.51	

→ No, *Hair length* and *Height* are not independent. Tall people have shorter hair!

Does that sound right to you?

→ Usually, random variables are independent only under particular conditions:

conditional independence, see next section.

Conditional Independence

Definition. Given sets of random variables $\mathbf{Z}_1$, $\mathbf{Z}_2$, $\mathbf{Z}$, we say that $\mathbf{Z}_1$ and $\mathbf{Z}_2$ are **conditionally independent given $\mathbf{Z}$** if:

$$\mathbf{P}(\mathbf{Z}_1, \mathbf{Z}_2 \mid \mathbf{Z}) = \mathbf{P}(\mathbf{Z}_1 \mid \mathbf{Z})\mathbf{P}(\mathbf{Z}_2 \mid \mathbf{Z})$$

We alternatively say that $\mathbf{Z}_1$ is **conditionally independent of $\mathbf{Z}_2$ given $\mathbf{Z}$** .

$P(\text{Hair length, Height} \mid \text{Sex} = \text{female})$, $P(\text{Height} \mid \text{Sex} = \text{female})$,
 $P(\text{Height} \mid \text{Hair length, Sex} = \text{female})$:

Height	Hair length		
	long	short	
tall	0.14	0.06	0.2
	0.2	0.2	
medium	0.56	0.24	0.8
	0.8	0.8	

$\leadsto$ Hair length and Height are independent given Sex=female.

Also given Sex=male, so Hair length and Height are independent given Sex.

Proposition. If $\mathbf{Z}_1$ and $\mathbf{Z}_2$ are conditionally independent given $\mathbf{Z}$, then
 $\mathbf{P}(\mathbf{Z}_1 \mid \mathbf{Z}_2, \mathbf{Z}) = \mathbf{P}(\mathbf{Z}_1 \mid \mathbf{Z})$.

If we already know the Sex of a person, knowing the hair length does not tell us anything about their height!

John, Mary, and My Brand-New Alarm

Example

I got very valuable stuff at home. So I bought an alarm. Unfortunately, the alarm just rings at home, doesn't call me on my mobile. I've got two neighbors, Mary and John, who'll call me if they hear the alarm. The problem is that, sometimes, the alarm is caused by an earthquake. Also, John might confuse the alarm with his telephone, and Maria might miss the alarm altogether because she typically listens to loud music.

Random variables: (All Boolean)

Burglary, Earthquake, Alarm, JohnCalls, MaryCalls

Question: We want to compute the probability of atomic events (e.g. $P(\text{Burglary}, \text{Earthquake}, \text{Alarm}, \text{JohnCalls}, \text{MaryCalls})$).
Do we need to store a table with 2^5 combinations?

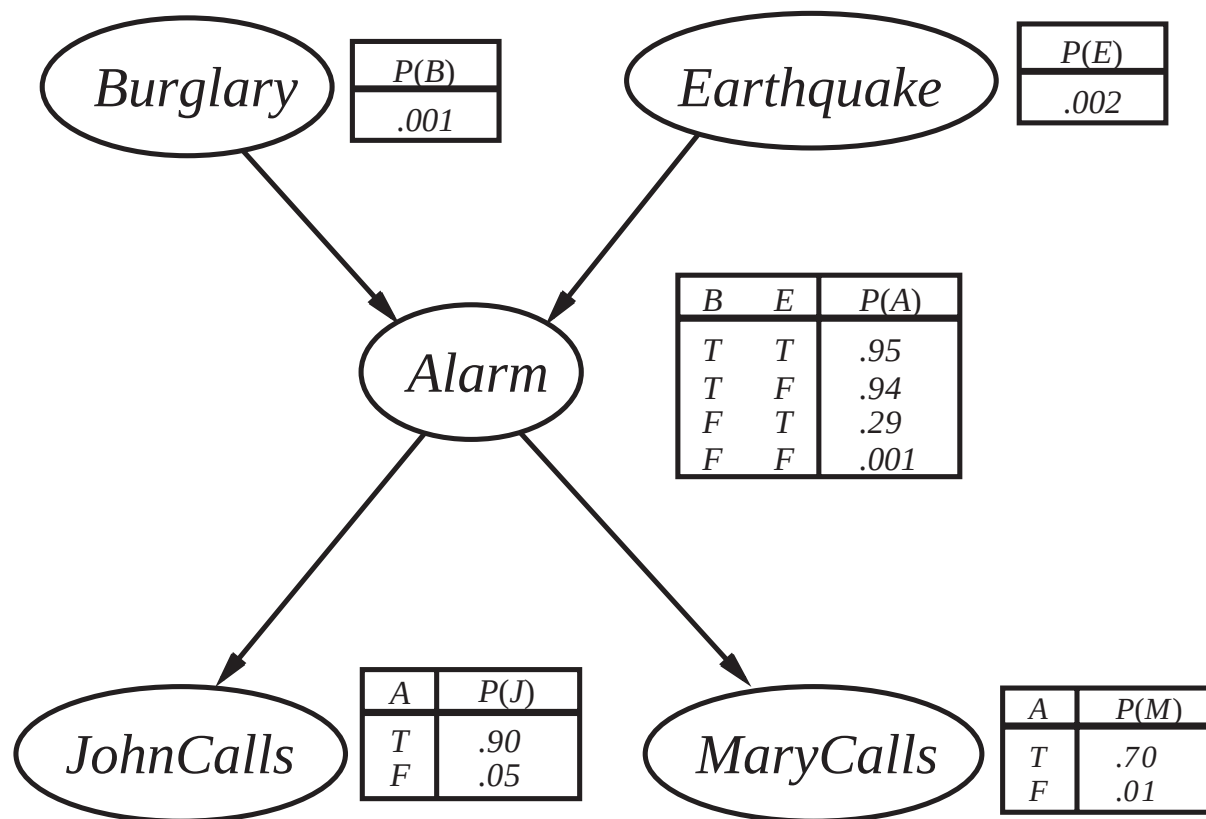
John, Mary, and My Alarm: Designing the BN

Cooking Recipe: (1) Design the random variables $X_1, \dots, X_n$; (2) Identify their dependencies; (3) Insert the conditional probability tables $\mathbf{P}(X_i \mid \text{Parents}(X_i))$.

Example: Let's cook!

- ① **Random variables:** *Burglary, Earthquake, Alarm, JohnCalls, MaryCalls*. (All Boolean)
- ② **Dependencies:** Burglaries and earthquakes are independent (this is actually debatable $\rightarrow$ design decision!); the alarm might be activated by either. John and Mary call if and only if they hear the alarm (they don't care about earthquakes).
- ③ **Conditional probability tables:** Assess the probabilities, see next slide.

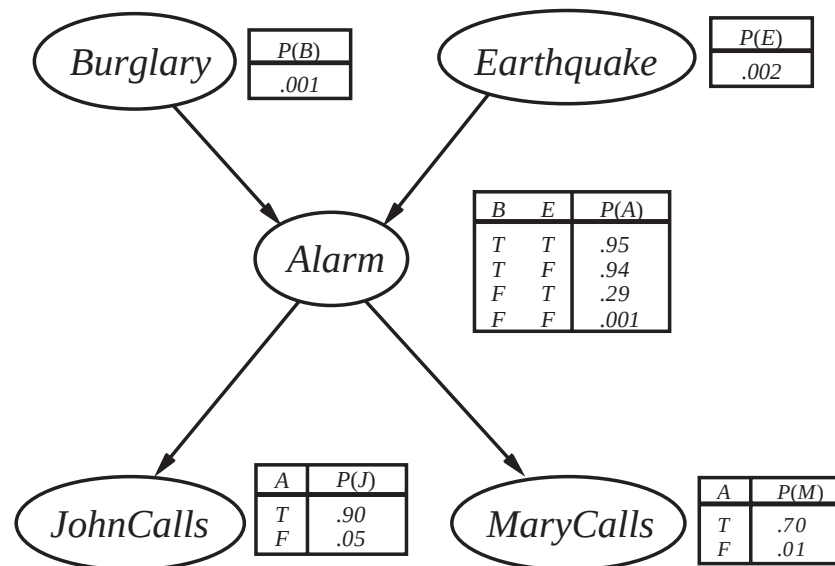
John, Mary, and My Alarm: The BN



Note: In each $\mathbf{P}(X_i \mid \text{Parents}(X_i))$, we show only $\mathbf{P}(X_i = \text{true} \mid \text{Parents}(X_i))$. We don't show $\mathbf{P}(X_i = \text{false} \mid \text{Parents}(X_i))$ which is $= 1 - \mathbf{P}(X_i = \text{true} \mid \text{Parents}(X_i))$.

The Syntax of Bayesian Networks

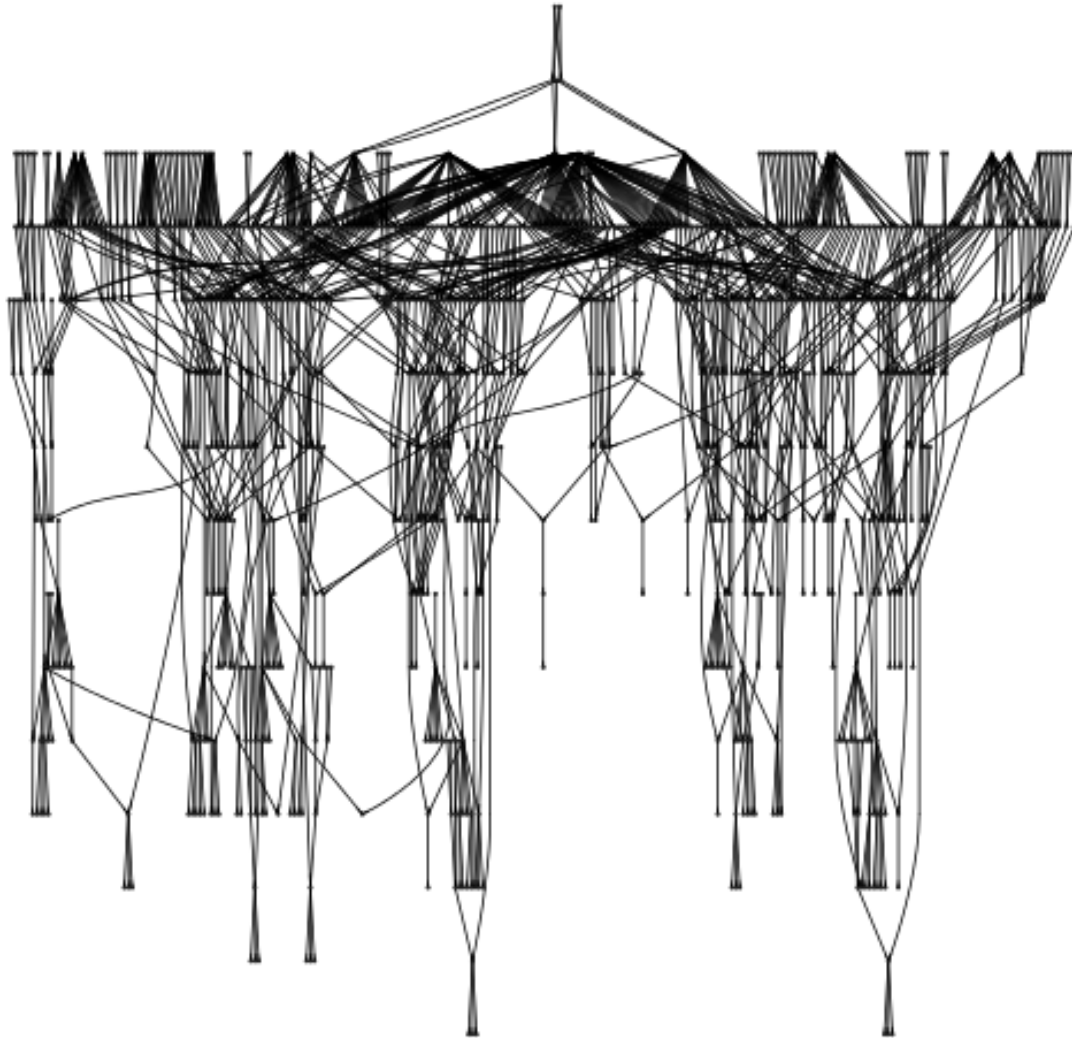
Representation of conditional dependencies in a **directed** and **acyclic** graph:



Definition (Bayesian Network). Given random variables $X_1, \dots, X_n$ with finite domains $D_1, \dots, D_n$, a **Bayesian network** is an acyclic directed graph $BN = (\{X_1, \dots, X_n\}, E)$. We denote $Parents(X_i) := \{X_j \mid (X_j, X_i) \in E\}$. Each X_i is associated with a function $CPT(X_i) : D_i \times (\prod_{X_j \in Parents(X_i)} D_j) \mapsto [0, 1]$. $CPT(X_i)$ is a **conditional probability table** specifying the conditional distribution $P(X_i \mid parents(A_i))$.

[$\rightarrow$ Why “acyclic”? Slide 19 (*) $\mathbf{P}(X_1, \dots, X_n) = \prod_{i=1}^n \mathbf{P}(X_i \mid Parents(X_i))$. By (*), acyclic BN suffice to represent any full joint probability distribution. But for cyclic BN, (*) does NOT hold, indeed cyclic BNs may be self-contradictory.]

The Munin network

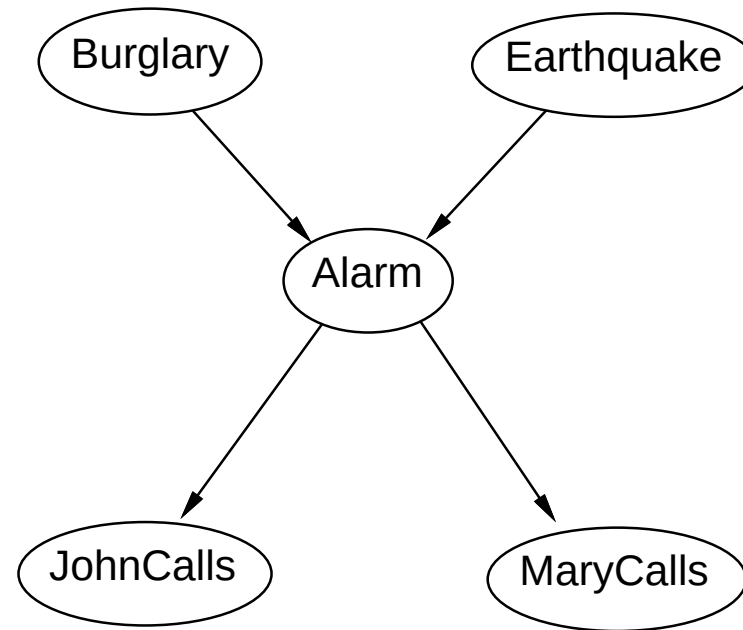


Characteristics:

- Approximately 1100 variables.
- Each variable has between 2 and 20 values.
- 10^{600} possible state configurations!

A system for diagnosing neuro-muscular diseases.

The Semantics of BNs: Example

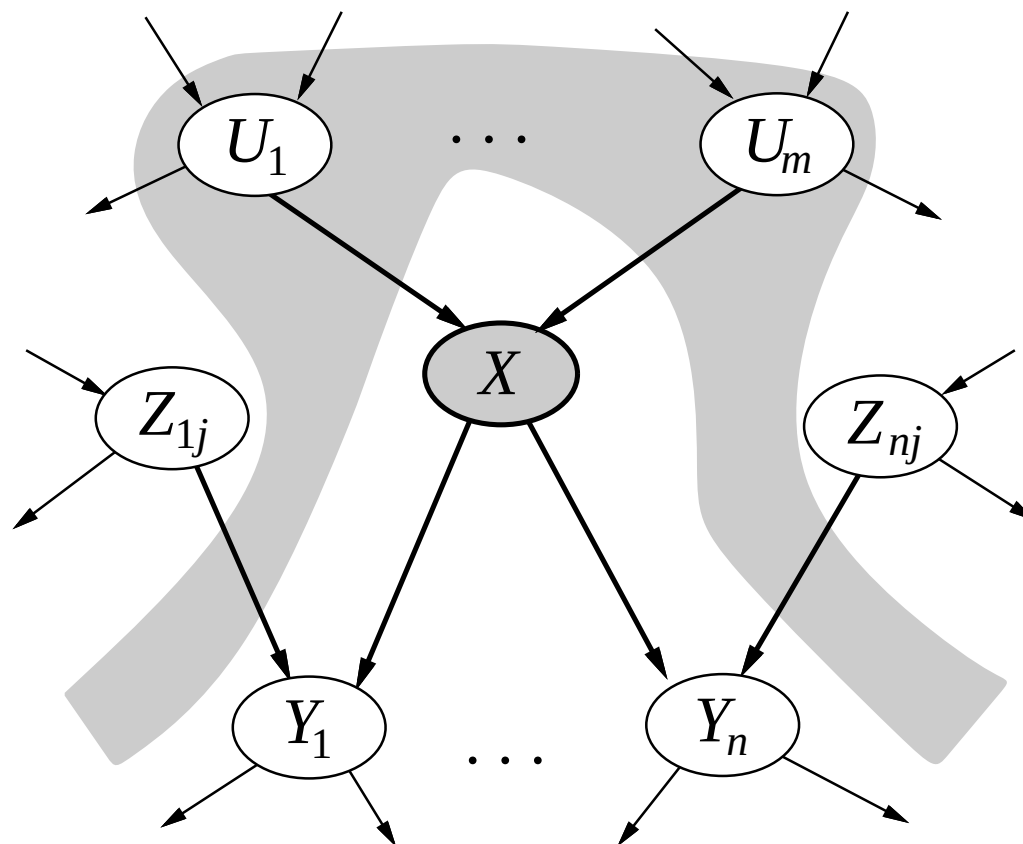


- *Alarm* depends on *Burglary* and *Earthquake*.
- *MaryCalls* only depends on *Alarm*.
$$P(\text{MaryCalls} \mid \text{Alarm}, \text{Burglary}) = P(\text{MaryCalls} \mid \text{Alarm})$$

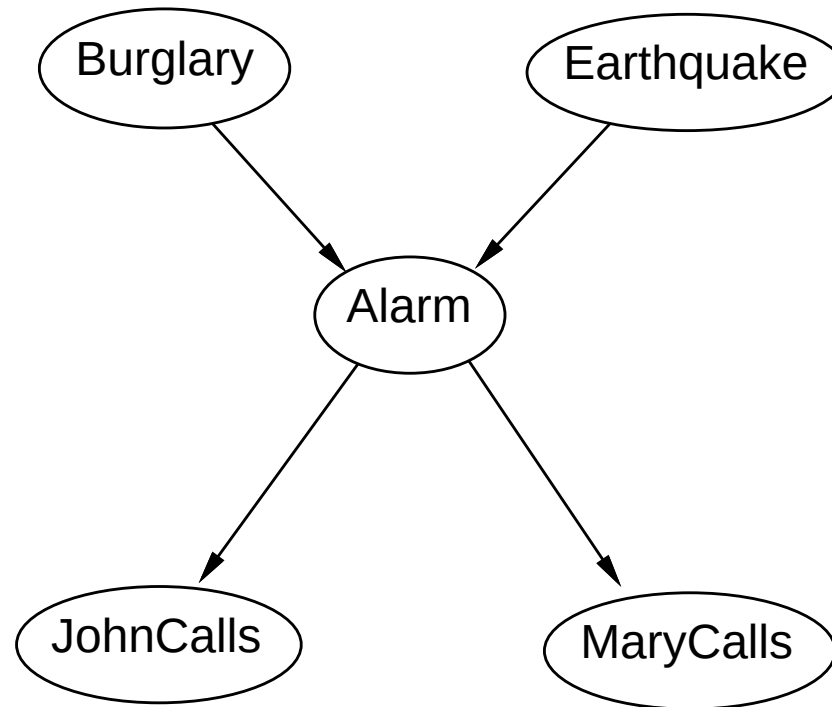
→ Bayesian networks represent sets of independence assumptions.

The Semantics of BNs: General Case

→ Each node X in a BN is conditionally independent of its **non-descendants** given its parents $Parents(X)$.

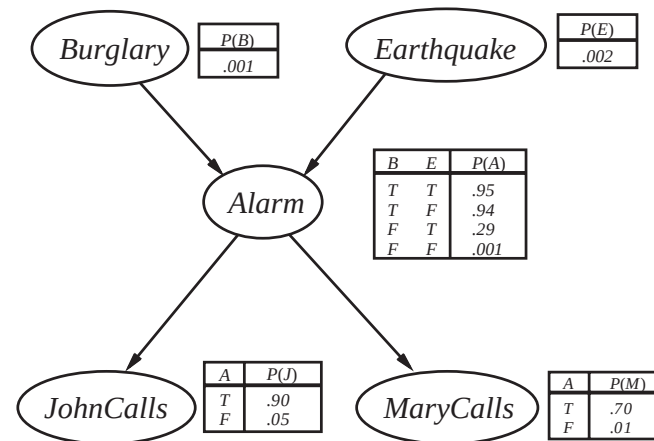


The Semantics of BNs: Example, ctd.



→ **Given the value of *Alarm*, *MaryCalls* is independent of?**
 $\{Burglary, Earthquake, JohnCalls\}$.

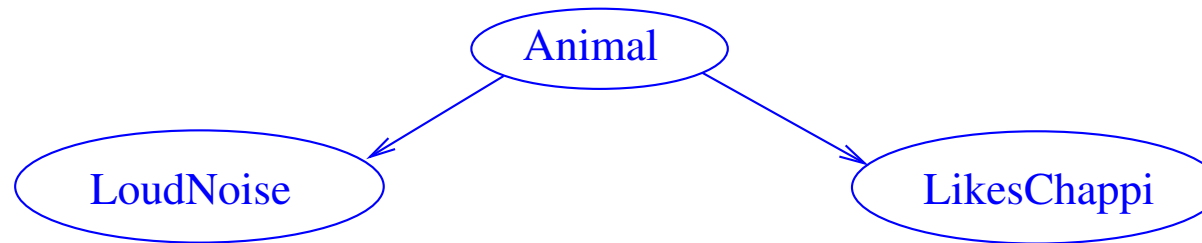
The Semantics of BNs: Formal



Definition. Given a Bayesian network $BN = (\{X_1, \dots, X_n\}, E)$, we identify BN with the following two assumptions:

- Ⓐ For $1 \leq i \leq n$, X_i is conditionally independent of $NonDescendants(X_i)$ given $Parents(X_i)$, where $NonDescendants(X_i) := \{X_j \mid (X_i, X_j) \notin E^*\} \setminus Parents(X_i)$ with E^* denoting the transitive closure of E .
- Ⓑ For $1 \leq i \leq n$, all values x_i of X_i , and all value combinations $parents(X_i)$ of $Parents(X_i)$, we have $P(x_i \mid parents(X_i)) = CPT(x_i, parents(X_i))$.

Questionnaire



Question!

Say BN is the Bayesian network above. Which statements are correct?

(A): *Animal is independent of LikesChappi.*

(B): *LoudNoise is independent of LikesChappi.*

(C): *Animal is conditionally independent of LikesChappi given LoudNoise.*

(D): *LikesChappi is conditionally independent of LoudNoise given Animal.*

(A) No: *likeschappi* indicates *dog*.

(B) No: Not knowing what animal it is, *likeschappi* is an indication for *dog* which indicates *loudnoise*.

(C) No: For example, even if we know *loudnoise*, knowing in addition that *likeschappi* gives us a stronger indication of $Animal = dog$.

(D) Yes: $X_i = LikesChappi$ is conditionally independent of $NonDescendants(X_i) = \{LoudNoise\}$ given $Parents(X_i) = \{Animal\}$.

The Product Rule

Proposition (Product Rule). Given propositions A and B , $P(a \wedge b) = P(a \mid b)P(b)$.
(Direct from definition.)

Example: $P(\text{cavity} \wedge \text{toothache}) = P(\text{toothache} \mid \text{cavity})P(\text{cavity})$.

→ If we know the values of $P(a \mid b)$ and $P(b)$, then we can compute $P(a \wedge b)$.

→ Similarly, $P(a \wedge b) = P(b \mid a)P(a)$.

Notation: $\mathbf{P}(X, Y) = \mathbf{P}(X \mid Y)\mathbf{P}(Y)$ is a **system of equations**:

$$P(W = \text{sunny} \wedge \text{headache}) = P(W = \text{sunny} \mid \text{headache})P(\text{headache})$$

$$P(W = \text{rain} \wedge \text{headache}) = P(W = \text{rain} \mid \text{headache})P(\text{headache})$$

$$\dots = \dots$$

$$P(W = \text{snow} \wedge \neg \text{headache}) = P(W = \text{snow} \mid \neg \text{headache})P(\neg \text{headache})$$

→ Similar for unconditional distributions, $\mathbf{P}(X, Y) = \mathbf{P}(X)\mathbf{P}(Y)$.

The Chain Rule

Proposition (Chain Rule). *Given random variables $X_1, \dots, X_n$, we have*

$$\mathbf{P}(X_1, \dots, X_n) = \mathbf{P}(X_n \mid X_{n-1}, \dots, X_1) * \mathbf{P}(X_{n-1} \mid X_{n-2}, \dots, X_1) * \dots * \mathbf{P}(X_2 \mid X_1) * \mathbf{P}(X_1).$$

Example: $P(\neg brush \wedge cavity \wedge toothache)$
 $= P(toothache \mid cavity, \neg brush)P(cavity, \neg brush)$
 $= P(toothache \mid cavity, \neg brush)P(cavity \mid \neg brush)P(\neg brush).$

Proof. Iterated application of Product Rule. $\mathbf{P}(X_1, \dots, X_n) = \mathbf{P}(X_n \mid X_{n-1}, \dots, X_1) * \mathbf{P}(X_{n-1}, \dots, X_1)$ by Product Rule. In turn, $\mathbf{P}(X_{n-1}, \dots, X_1) = \mathbf{P}(X_{n-1} \mid X_{n-2}, \dots, X_1) * \mathbf{P}(X_{n-2}, \dots, X_1)$, etc.

Note: This works *for any ordering* of the variables.

→ We can recover the probability of atomic events from sequenced conditional probabilities for any ordering of the variables.

→ **First of the four basic techniques in Bayesian networks.**

Recovering the Full Joint Probability Distribution

“A Bayesian network is **a methodology for representing the full joint probability distribution.**”

→ How to recover the full joint probability distribution $\mathbf{P}(X_1, \dots, X_n)$ from $BN = (\{X_1, \dots, X_n\}, E)$?

Chain rule: For **any** ordering $X_1, \dots, X_n$, we have:

$$\mathbf{P}(\mathbf{X}_1, \dots, \mathbf{X}_n) = \mathbf{P}(\mathbf{X}_n \mid \mathbf{X}_{n-1}, \dots, \mathbf{X}_1) \mathbf{P}(\mathbf{X}_{n-1} \mid \mathbf{X}_{n-2}, \dots, \mathbf{X}_1) \dots \mathbf{P}(\mathbf{X}_1)$$

Choose $X_1, \dots, X_n$ **consistent with BN**: $X_j \in Parents(X_i) \implies j < i$.

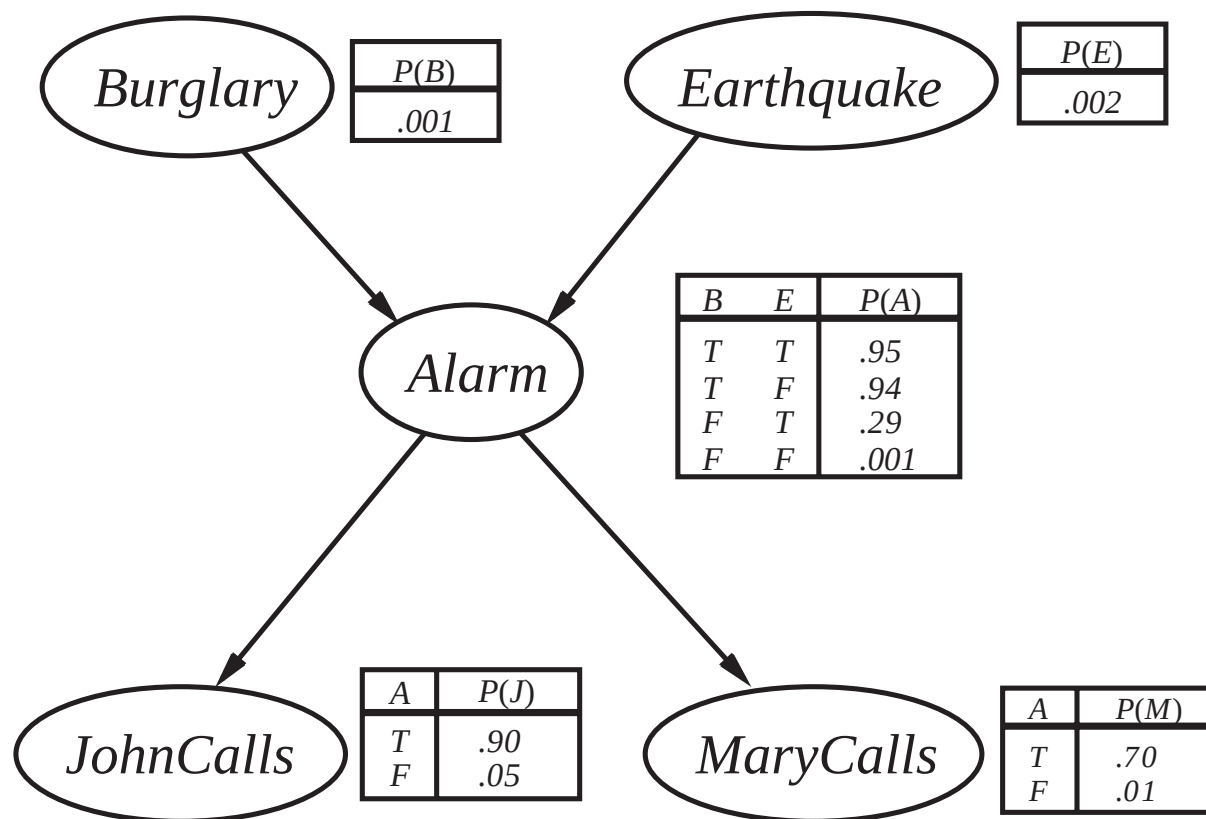
Exploit conditional independence: With **BN assumption (A)**, instead of $\mathbf{P}(X_i \mid X_{i-1} \dots, X_1)$ we can use $\mathbf{P}(X_i \mid Parents(X_i))$:

$$\mathbf{P}(\mathbf{X}_1, \dots, \mathbf{X}_n) = \prod_{i=1}^n \mathbf{P}(\mathbf{X}_i \mid Parents(\mathbf{X}_i))$$

The distributions $\mathbf{P}(X_i \mid Parents(X_i))$ are given by **BN assumption (B)**.

→ Same for atomic events $P(x_1, \dots, x_n)$.

Recovering a Probability for John, Mary, and the Alarm



$$\begin{aligned}
 \mathbf{P(j, m, a, \neg b, \neg e)} &= P(j \mid a) \cdot P(m \mid a) \cdot P(a \mid \neg b, \neg e) \cdot P(\neg b) \cdot P(\neg e) \\
 &= 0.9 * 0.7 * 0.001 * 0.999 * 0.998 \\
 &= 0.000628
 \end{aligned}$$

Example continued

Chain rule

$$\begin{aligned} P(\text{Burglary}, \text{Earthquake}, \text{Alarm}, \text{JohnCalls}, \text{MaryCalls}) = \\ P(\text{Burglary}) \cdot P(\text{Earthquake} \mid \text{Burglary}) \cdot P(\text{Alarm} \mid \text{Burglary}, \text{Earthquake}) \cdot \\ P(\text{JohnCalls} \mid \text{Burglary}, \text{Earthquake}, \text{Alarm}) \cdot \\ P(\text{MaryCalls} \mid \text{Burglary}, \text{Earthquake}, \text{Alarm}, \text{JohnCalls}) \end{aligned}$$

Conditional independence assumptions

$$P(\text{Earthquake} \mid \text{Burglary}) = P(\text{Earthquake})$$

Burglaries and earthquakes are independent (this is actually debatable → design decision!)

$$P(\text{Alarm} \mid \text{Burglary}, \text{Earthquake}) = P(\text{Alarm} \mid \text{Burglary}, \text{Earthquake})$$

John and Mary call if and only if they hear the alarm (they don't care about earthquakes)

$$P(\text{JohnCalls} \mid \text{Burglary}, \text{Earthquake}, \text{Alarm}) = P(\text{JohnCalls} \mid \text{Alarm})$$

$$P(\text{MaryCalls} \mid \text{Burglary}, \text{Earthquake}, \text{Alarm}, \text{JohnCalls}) = P(\text{MaryCalls} \mid \text{Alarm})$$

Simplified representation of joint distribution

$$\begin{aligned} P(\text{Burglary}, \text{Earthquake}, \text{Alarm}, \text{JohnCalls}, \text{MaryCalls}) = P(\text{Burglary}) \cdot P(\text{Earthquake}) \cdot \\ P(\text{Alarm} \mid \text{Burglary}, \text{Earthquake}) \cdot P(\text{JohnCalls} \mid \text{Alarm}) \cdot P(\text{MaryCalls} \mid \text{Alarm}) \end{aligned}$$

Constructing Bayesian Networks

BN construction algorithm:

1. Initialize $BN := (\{X_1, \dots, X_n\}, E)$ where $E = \emptyset$.
2. Fix any order of the variables, $X_1, \dots, X_n$. **Hint: causes before effects**
3. **for** $i := 1, \dots, n$ **do**
 - a. Choose **a minimal set** $Parents(X_i) \subseteq \{X_1, \dots, X_{i-1}\}$ **so that**
 $\mathbf{P}(X_i \mid X_{i-1} \dots, X_1) = \mathbf{P}(X_i \mid Parents(X_i))$.
 - b. For each $X_j \in Parents(X_i)$, insert (X_j, X_i) into E .
 - c. Associate X_i with $CPT(X_i)$ corresponding to $\mathbf{P}(X_i \mid Parents(X_i))$.

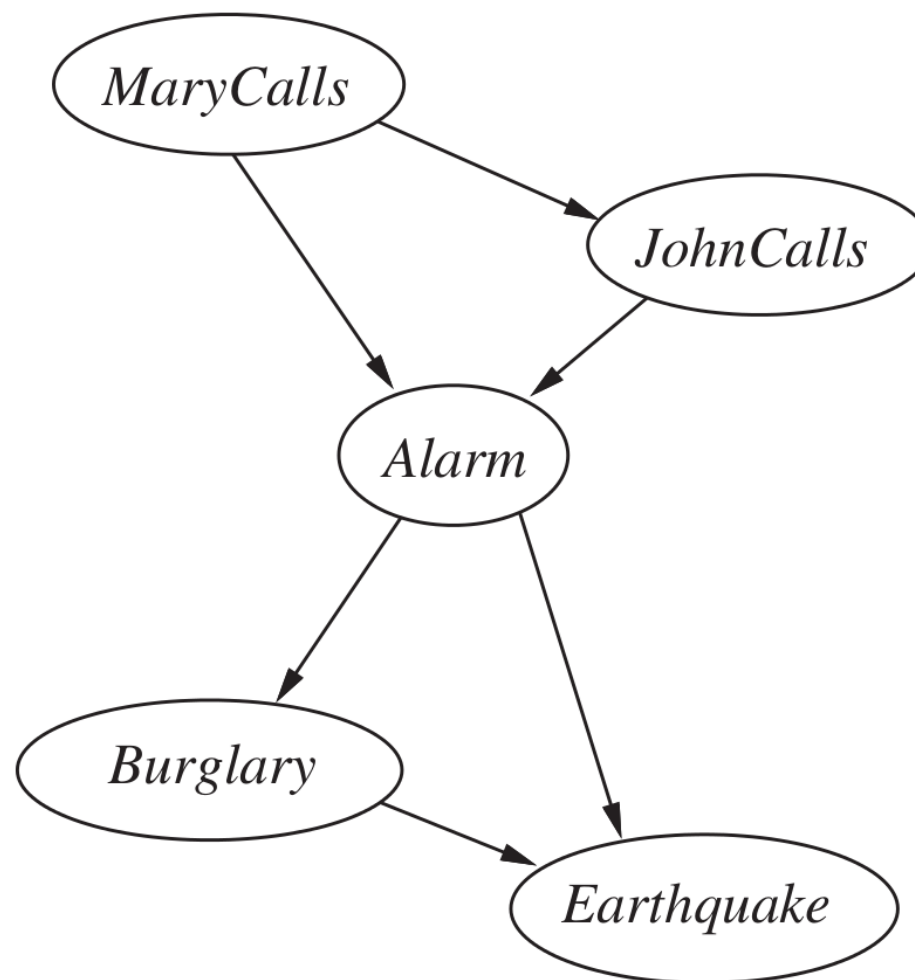
Attention! Which variables we need to include into $Parents(X_i)$ depends on what “ $\{X_1, \dots, X_{i-1}\}$ ” is ... !

→ **The size of the resulting BN depends on the chosen order $X_1, \dots, X_n$.**

→ The size of a Bayesian network is *not* a fixed property of the domain. It depends on the skill of the designer.

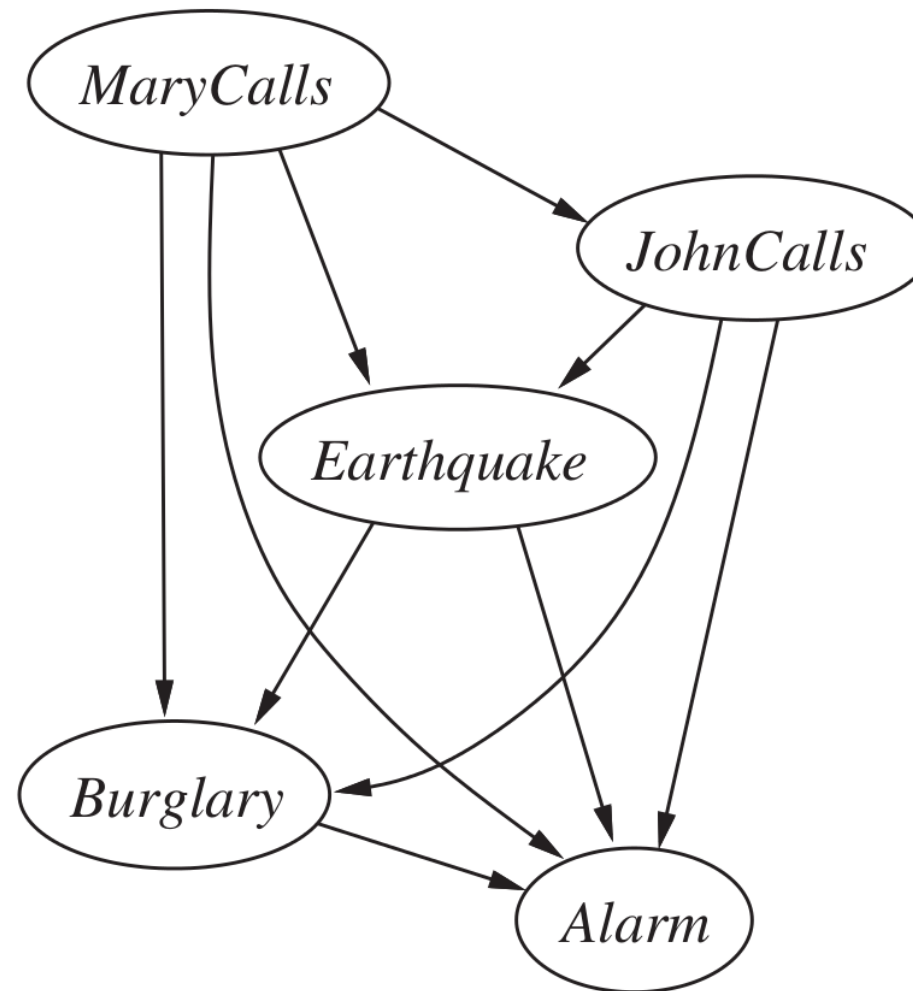
John and Mary Depend on the Variable Order!

Example: *MaryCalls*, *JohnCalls*, *Alarm*, *Burglary*, *Earthquake*.

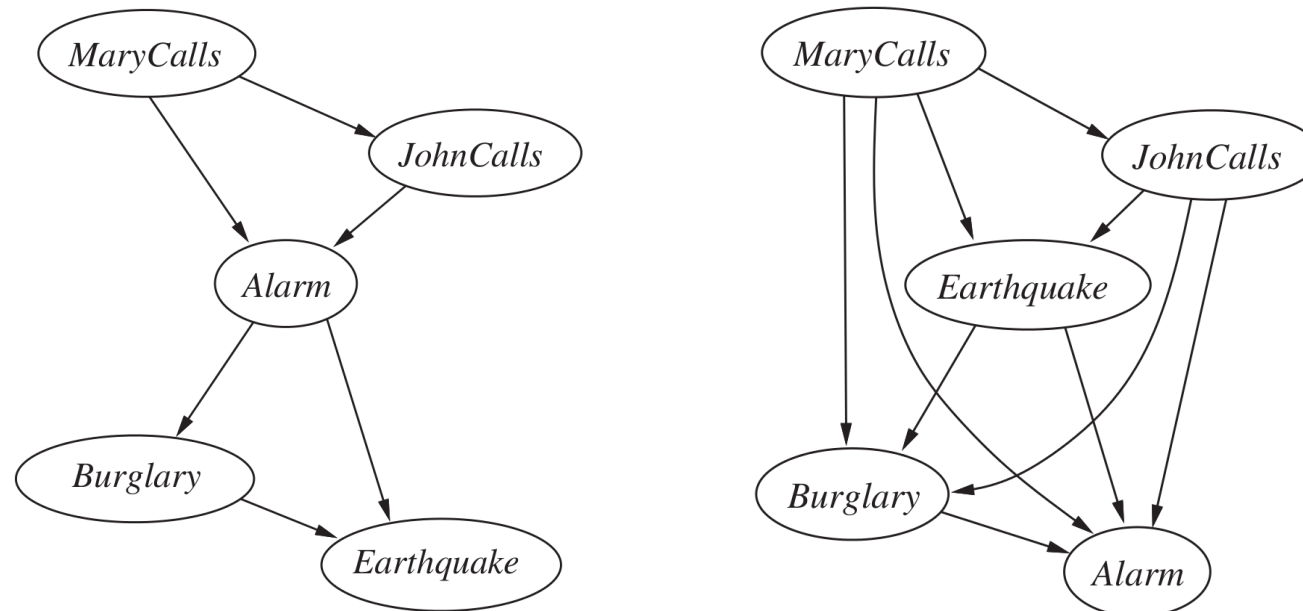


John and Mary Depend on the Variable Order! Ctd.

Example: *MaryCalls*, *JohnCalls*, *Earthquake*, *Burglary*, *Alarm*.



John and Mary, What Went Wrong?



→ **These BNs link from symptoms to causes!** ($\mathbf{P}(Cavity \mid Toothache)$)

- We fail to identify many conditional independence relations (e.g., get dependencies between conditionally independent symptoms).
- Also recall: Conditional probabilities $\mathbf{P}(Symptom \mid Cause)$ are more robust and often easier to assess than $\mathbf{P}(Cause \mid Symptom)$.

→ We should order causes before symptoms.

The Size of Bayesian Networks

Definition. Given random variables $X_1, \dots, X_n$ with finite domains $D_1, \dots, D_n$, the size of $BN = (\{X_1, \dots, X_n\}, E)$ is defined as $size(BN) := \sum_{i=1}^n |D_i| * \prod_{X_j \in Parents(X_i)} |D_j|$. (= The total number of entries in the CPTs.)

→ **Smaller BN $\implies$ assess less probabilities, more efficient inference.**

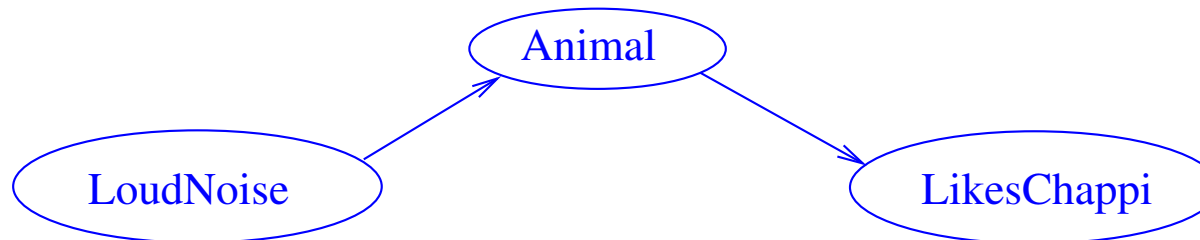
- Explicit full joint probability distribution has size $\prod_{i=1}^n |D_i|$.
- If $|Parents(X_i)| \leq k$ for every X_i , and $D_{\max}$ is the largest variable domain, then $size(BN) \leq n * |D_{\max}|^{k+1}$.
→ For $|D_{\max}| = 2$, $n = 20$, $k = 4$ we have $2^{20} = 1048576$ probabilities, but a Bayesian network of size $\leq 20 * 2^5 = 640 \dots!$
- In the worst case, $size(BN) = \sum_{i=1}^n \prod_{j=1}^i |D_j|$, namely if every variable depends on all its predecessors in the chosen order.

→ BNs are compact if each variable is directly influenced only by few of its predecessor variables.

Questionnaire

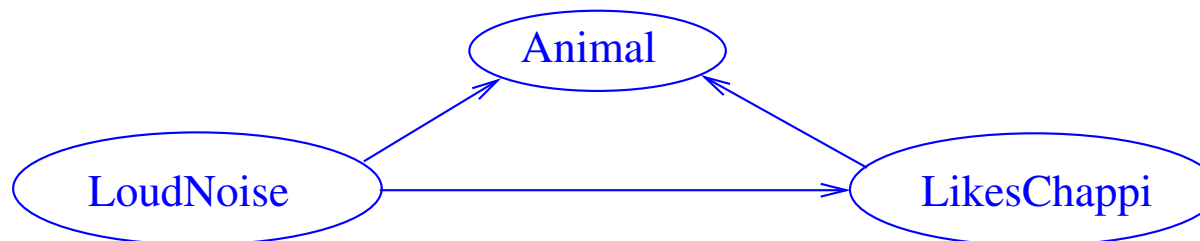
Question!

What is the Bayesian network we get by constructing according to the ordering $X_1 = LoudNoise$, $X_2 = Animal$, $X_3 = LikesChappi$?



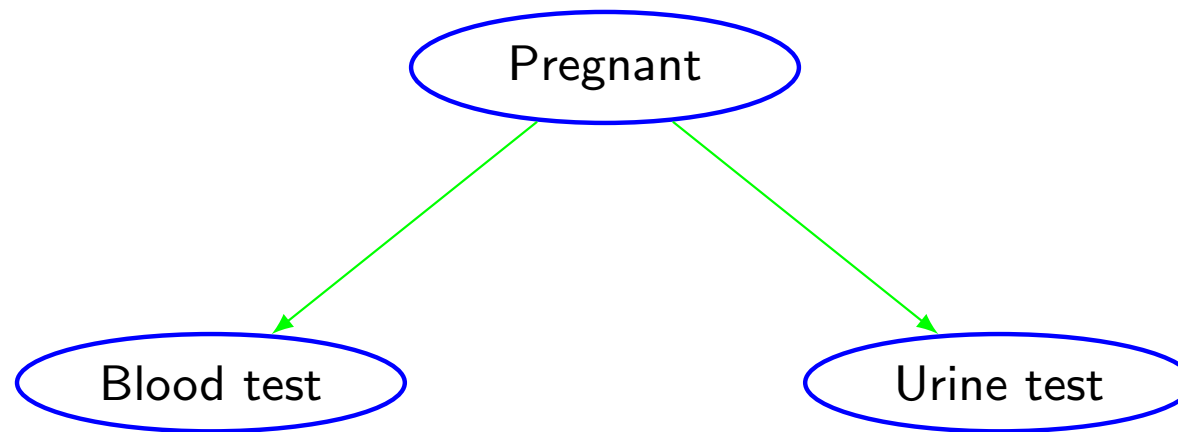
Question!

What is the Bayesian network we get by constructing according to the ordering $X_1 = LoudNoise$, $X_2 = LikesChappi$, $X_3 = Animal$?



Insemination of a cow

Six weeks after the insemination of a cow, there are two tests: a **Blood test** and a **Urine test**.



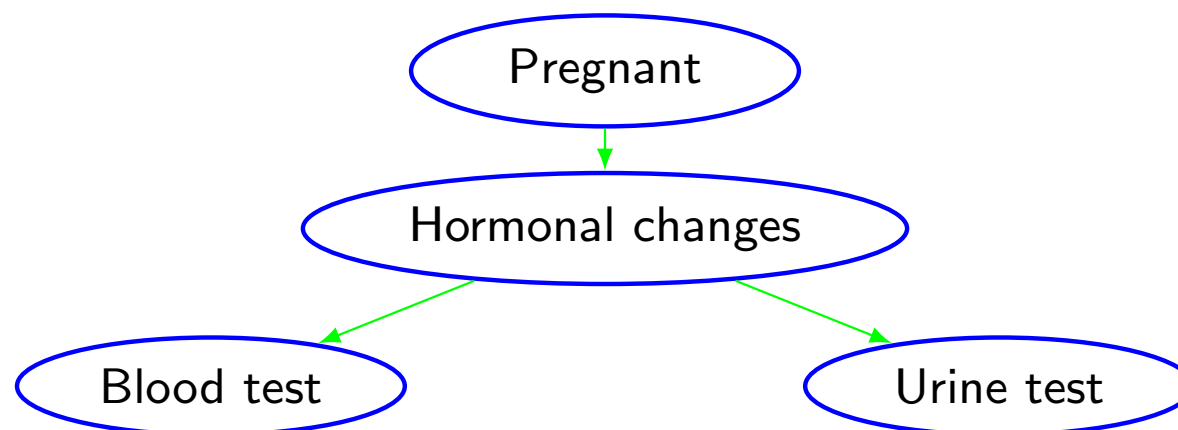
Check the conditional independences

If we know that the cow is pregnant, will a negative blood test then change our expectation for the urine test?

If **it will**, then the model does not reflect reality!

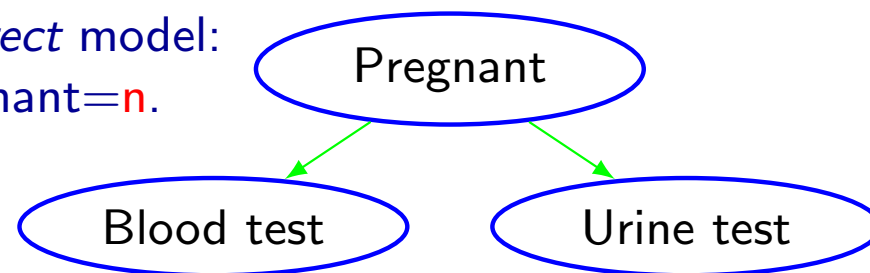
Insemination of a cow: A more correct model

We introduce a **mediating variable**: Hormonal changes



But does this actually make a difference?

Assume that both tests are **negative** in the *incorrect* model:
This will overestimate the probability for Pregnant=**n**.



Summary

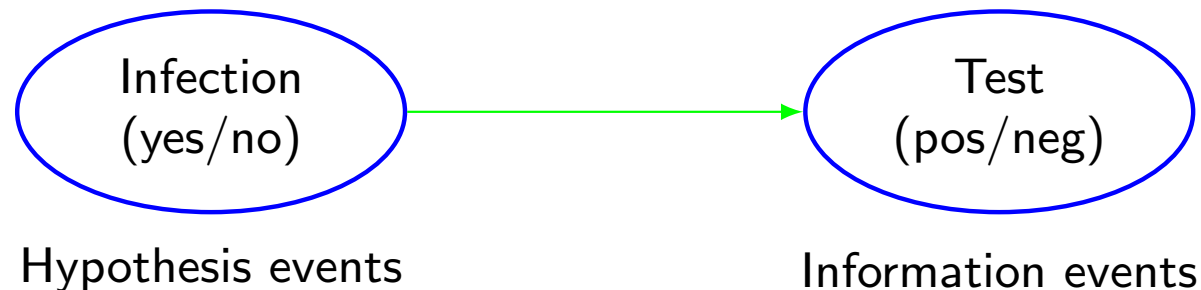
- **Reasoning** can be attained by a combination of **logic and probability**.
- **Probabilities** express the degree of belief about an event, given the agent's knowledge. **Conditional probabilities** express the likelihood of an event given observed evidence.
- **Assessing** a probability means to use statistics to approximate the likelihood of an event.
- **Bayes' rule** allows us to derive, from probabilities that are easy to assess, probabilities that aren't easy to assess.
- Given **multiple evidence**, we can exploit **conditional independence**.
- **Bayesian networks (BN)** are a wide-spread tool to model uncertainty, and to reason about it. A BN represents **conditional independence relations** between random variables. It consists of a graph encoding the variable dependencies, and of **conditional probability tables (CPTs)**.
- Given a BN we can recover the full joint probability distribution using the chain rule.
- Given a variable order, the BN is small if every variable depends on only a few of its predecessors.
- When designing Bayesian Networks, we need to consider what variables should be added and what are their dependencies. Introducing **mediating variables** can help to simplify the model.

Building models

Example

Milk from a cow may be infected. To detect whether or not the milk is infected, you can apply a test which may either give a positive or a negative test result. The test is not perfect: It may give **false positives** as well as **false negatives**.

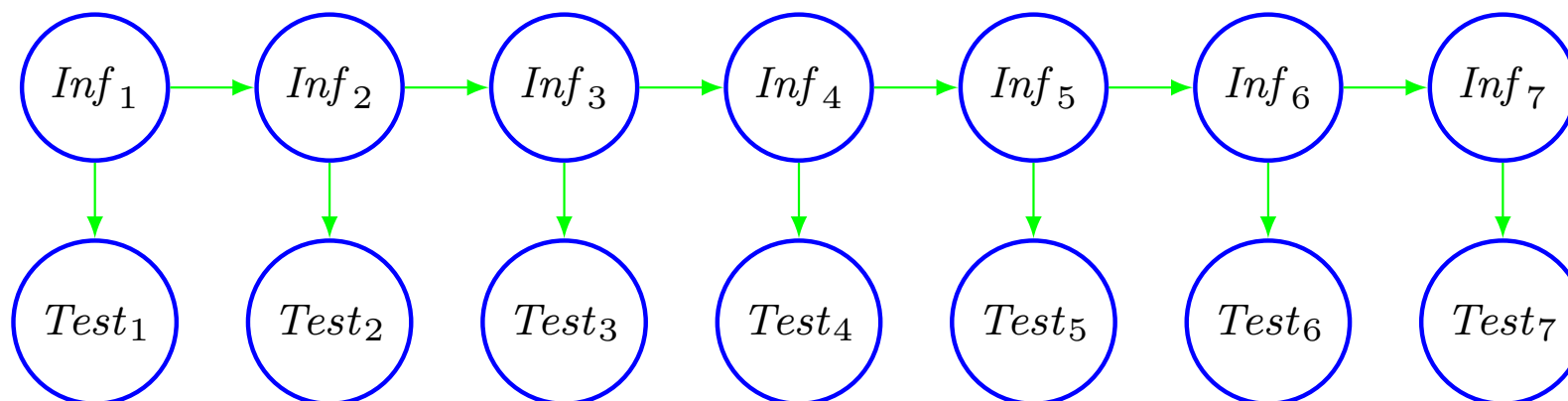
We set links from hypothesis to test (causal relationship: having the disease is what causes the test to be positive and not viceversa):



What if we repeat the test multiple days?

7-day model I

Infections develop over time:



Assumption

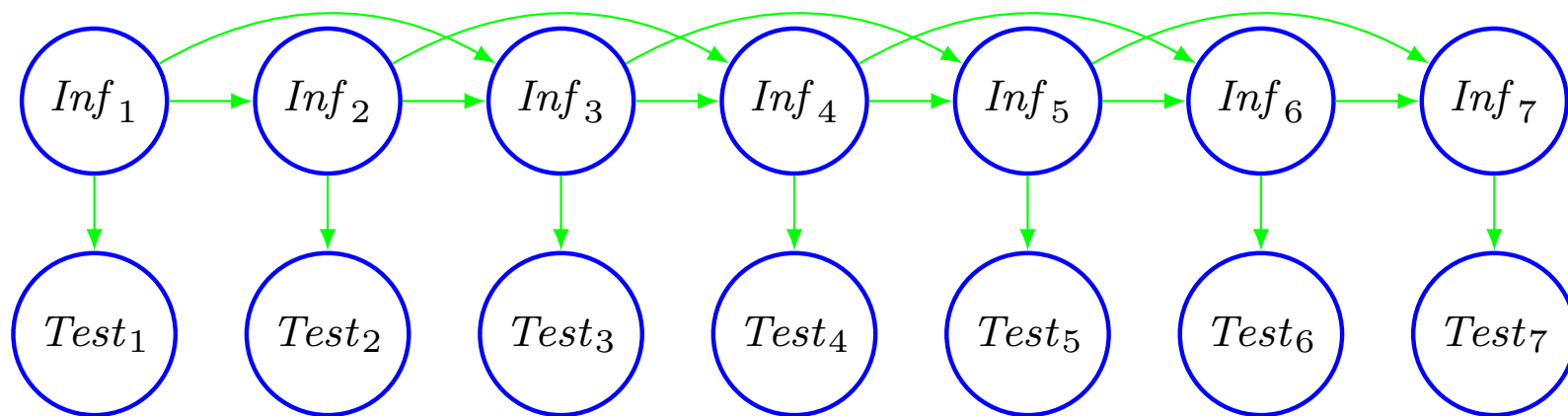
The **Markov property**: If I know the present, then the past has no influence on the future:

Inf_{i-1} is d-separated from Inf_{i+1} given Inf_i .

But what if yesterday's Inf-state has an impact on tomorrow's Inf-state?

7-day model II

Non-Markov relations



Yesterday's Inf-state has an impact on tomorrow's Inf-state.